National Blockchain Framework BaaS

User Manual | PART-B: SMART CONTRACTS

NBF Team
C-DAC | www.cdac.in

Revision History

Version	Date	Author	Remarks
0.1	24-05-2023	NBF Team	Initial document
0.2	25-03-2025	NBF Team	Updated template studio (Fabric)

Table of Contents

Revision History 1

1. National Blockchain Framework 2

2. Smart Contract layer 2

3. Template Library 2

4. Generic APIs 5

5. Template Studio 7

6.Admin functionalities 13

6.1 Register Domain 13

6.2 Add Stakeholders 19

6.3 Add Functions- 22

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1. National Blockchain Framework

The National Blockchain Framework is an initiative of the Ministry of Electronics and Information Technology (MeitY) to make India ready for large-scale adoption of Blockchain and enable trust for applications in the domain of e-Governance. The framework is being offered as a service with technology stack supporting the following capabilities:

- End-to-end, rapid and secure application development
- OpenAPIs for seamless integration
- Distributed infrastructure

The framework is also capable of hosting design patterns based on Smart Contract Templates addressing requirements of various application domains. This facilitates faster development and on-boarding of new and existing applications. It enables the Government departments to develop, operate and govern a Blockchain network and e-Governance application with the required performance and security.

BasS Features:

- Faster End-to-End Blockchain Stack Setup
- Single and Multi Node Setup
- Low Code Blockchain Development
- Application Domain specific and generic Smart Contracts
- Supported Platforms: Fabric and Sawtooth
- Security across Network, Smart Contract and Application layers
- Easy onboarding of new and existing applications
- OpenAPIs for seamless integration
- Bring Your Own Infrastructure
- Distributed infrastructure

This document helps the users to understand the complete functionality and features of BaaS.

2. Smart Contract layer

Smart contract layer consists of three utilities:

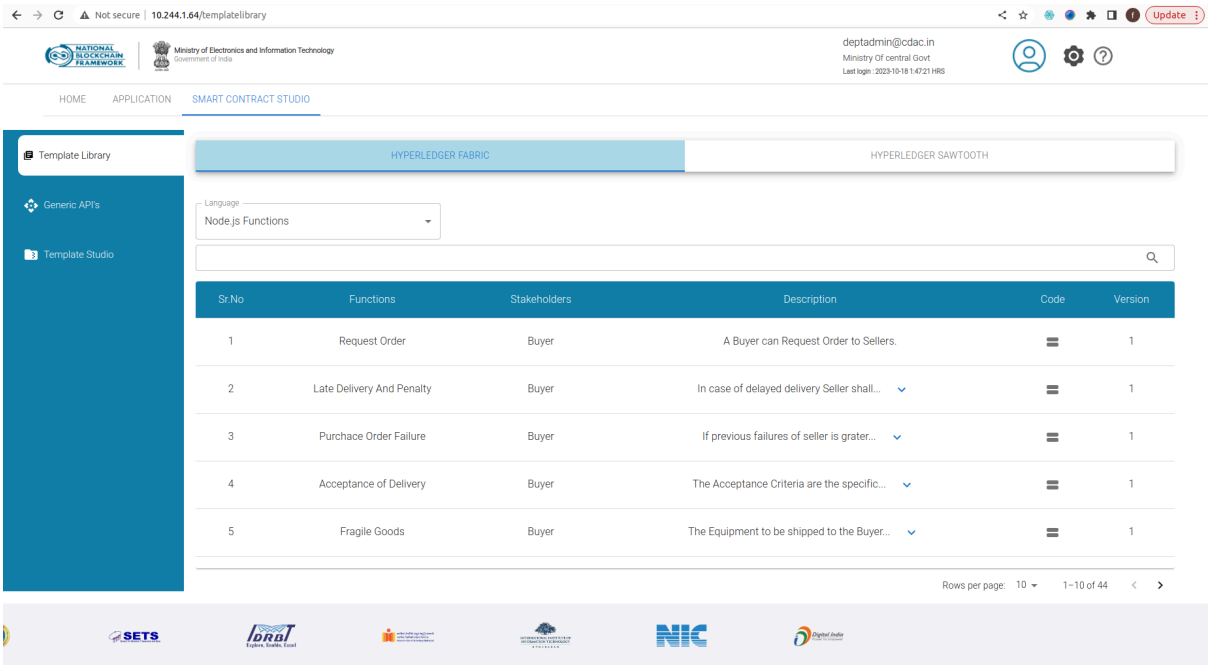
i. Template Library
ii. Generic APIs
iii. Template Studio

3. Template Library

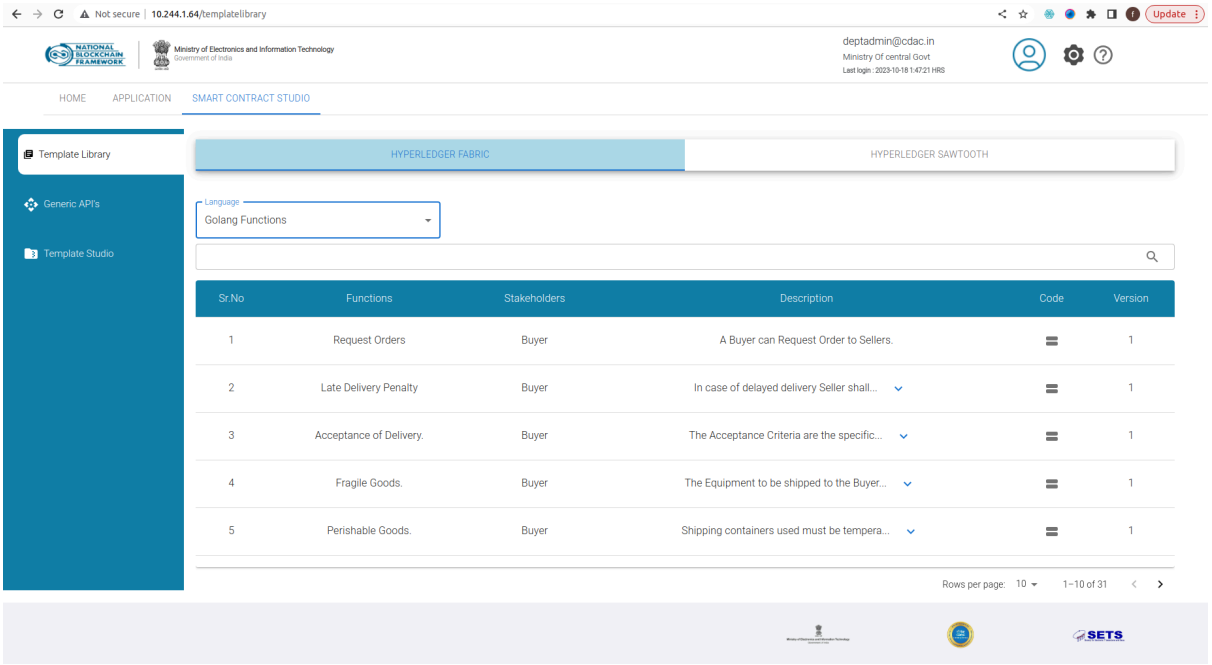
- Under the template library, users can select for NODE functions or GO functions from dropdown.The added functions,stakeholders, description, code and version are displayed here. The description of the function can be expanded and The code can also be viewed.

NODE FUNCTIONS:

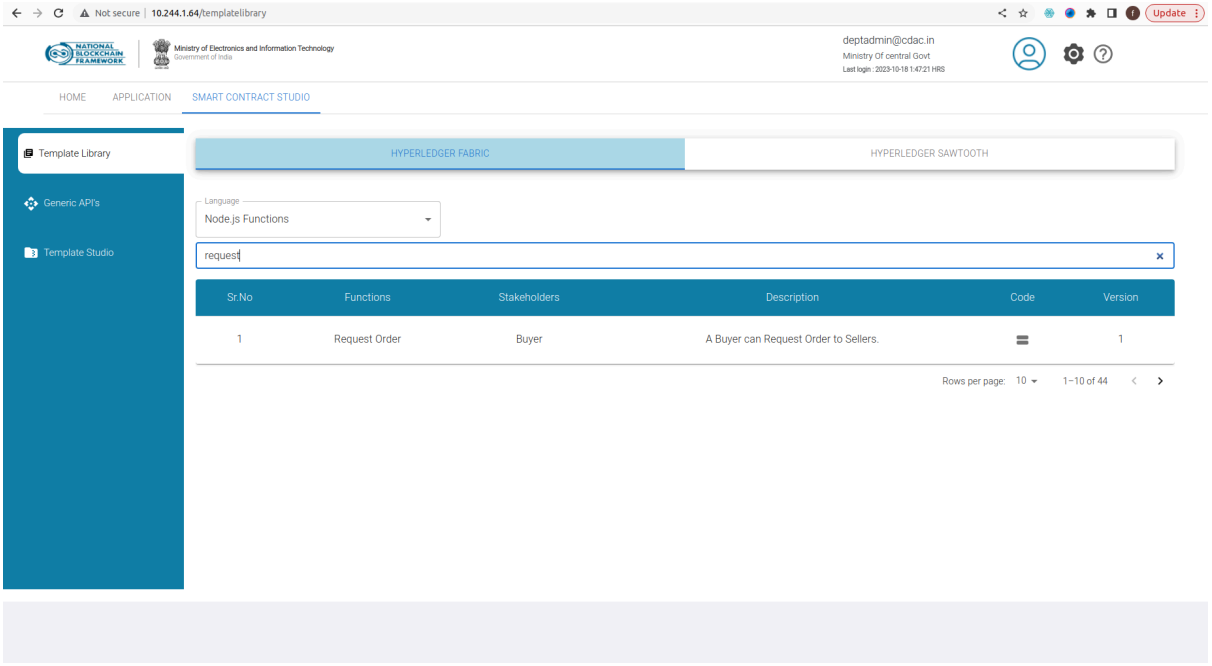
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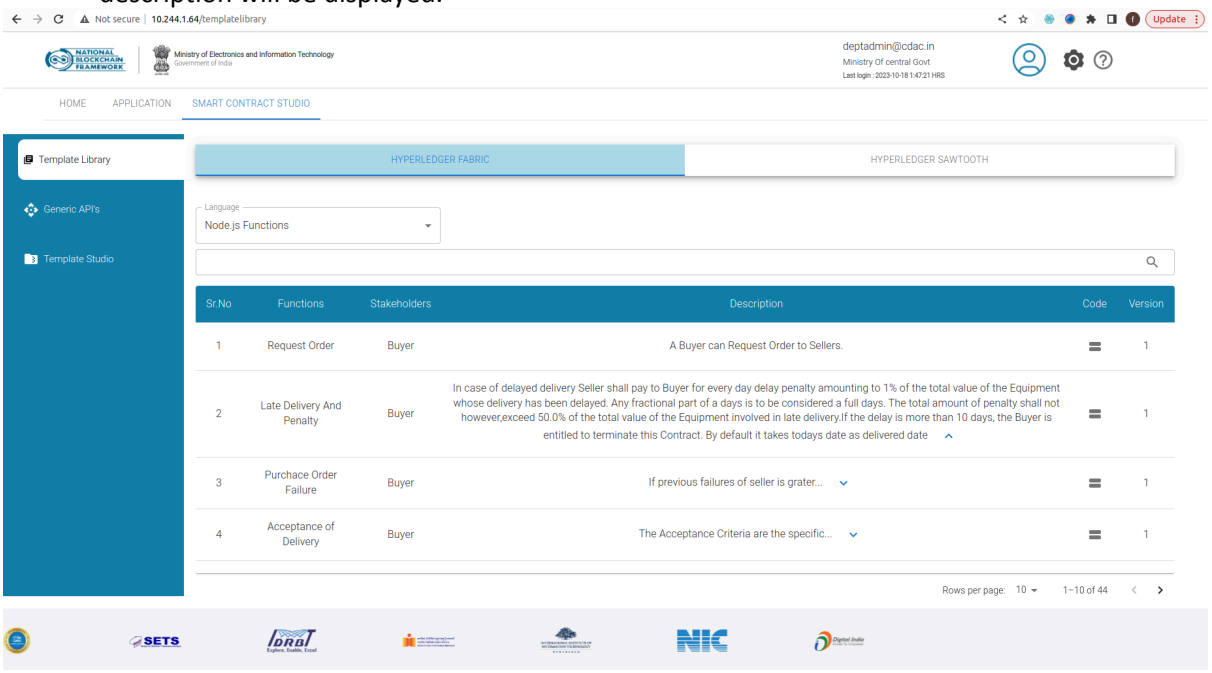
GO FUNCTIONS:



- For the users search facility is also provided. Users can search on the basis of functions ,description.



- To know more description about functions, users can click on the arrow mark and a detailed description will be displayed.

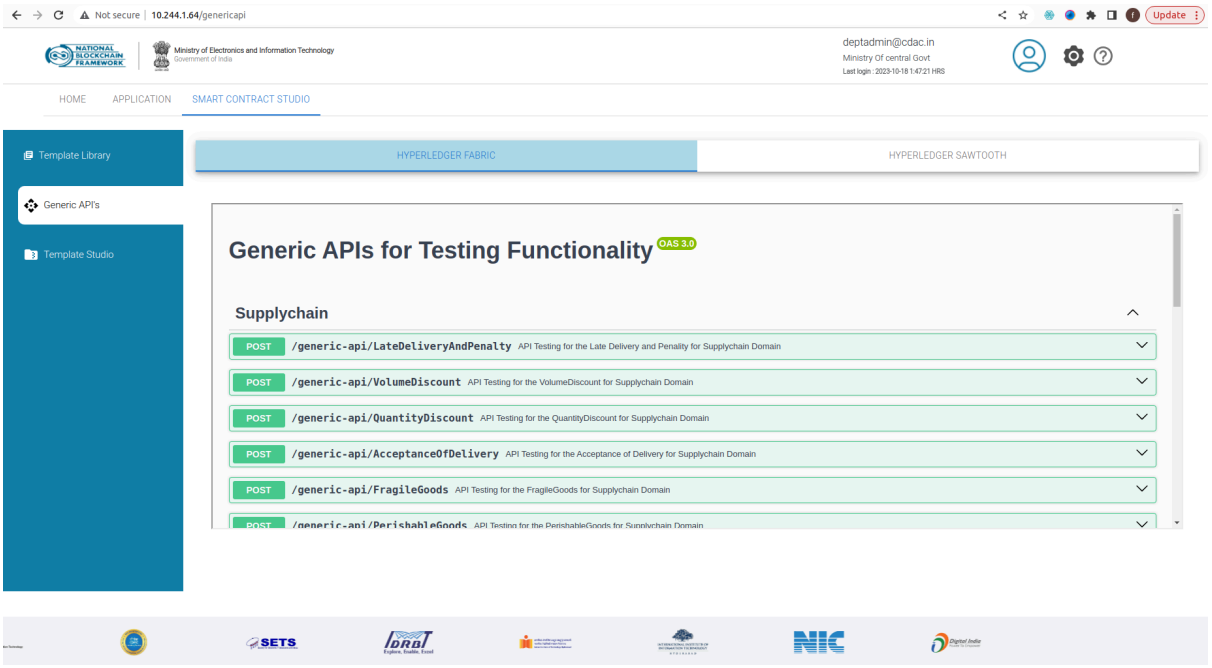


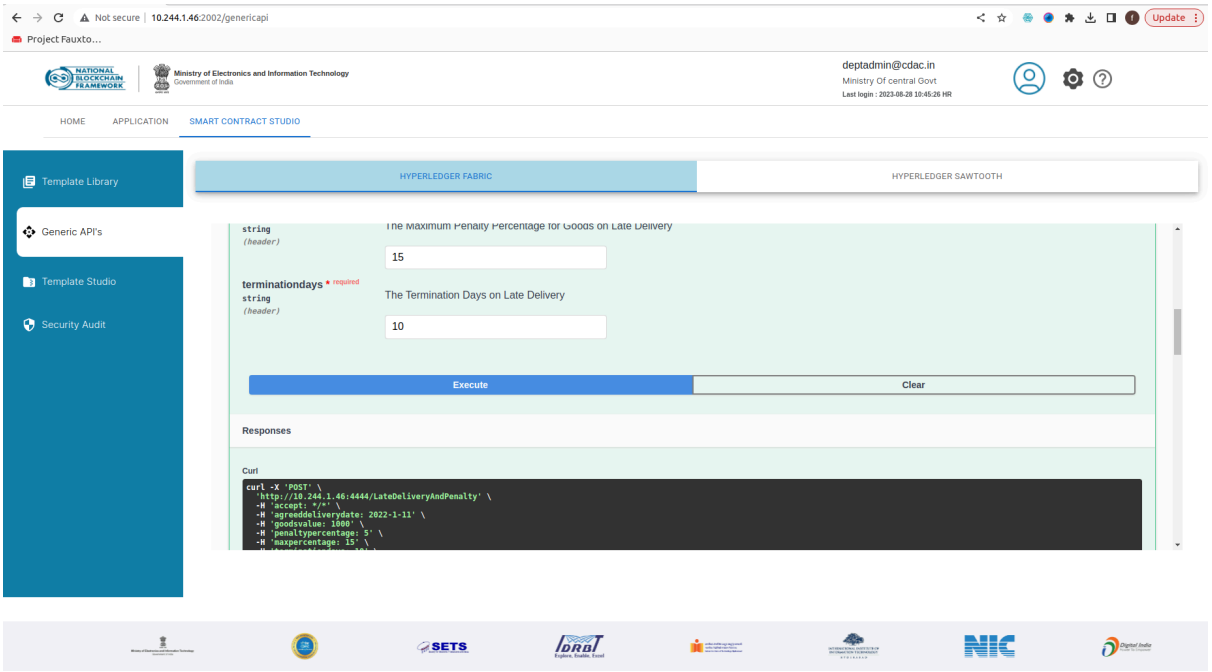
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4. Generic APIs

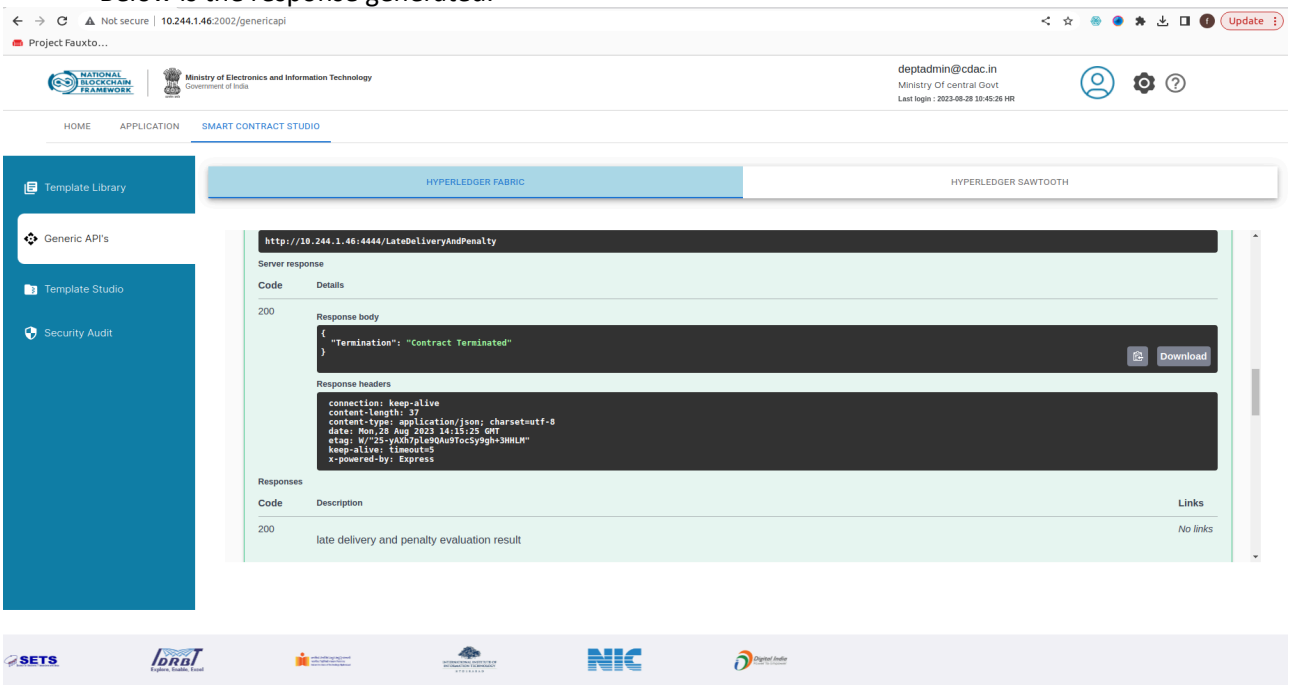
- Under the Generic APIs tab,users can test different functions of different domains by clicking on the “Try it out”button before bundling the chaincode by providing input parameters and then hitting the execute button.

- After that a response will be generated based on the given parameters.



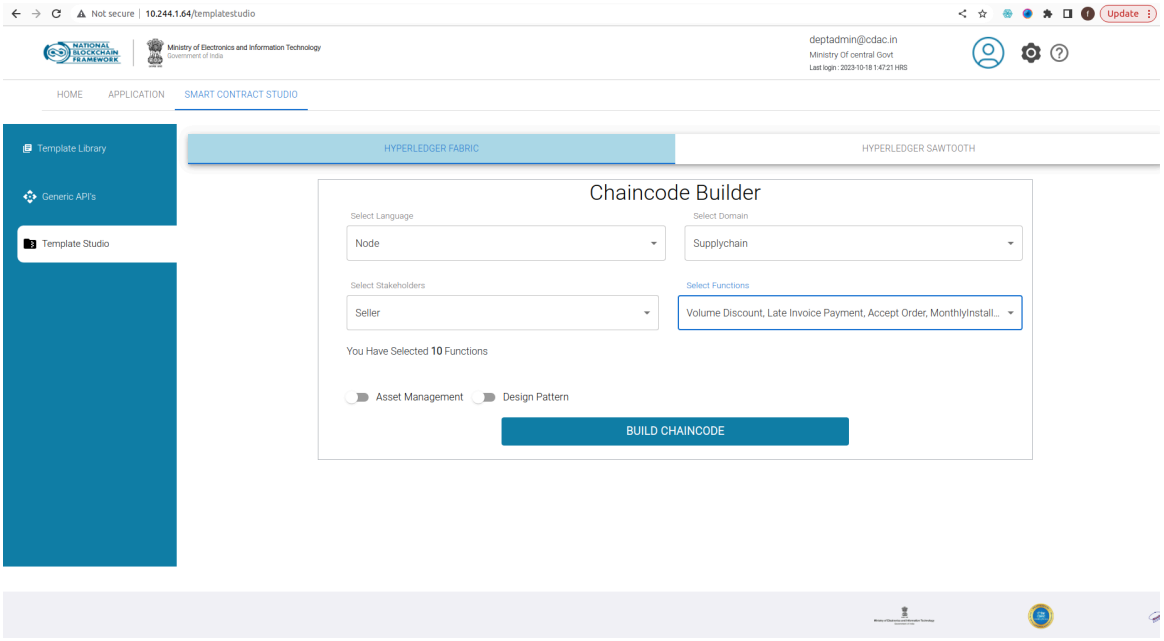


- Below is the response generated.

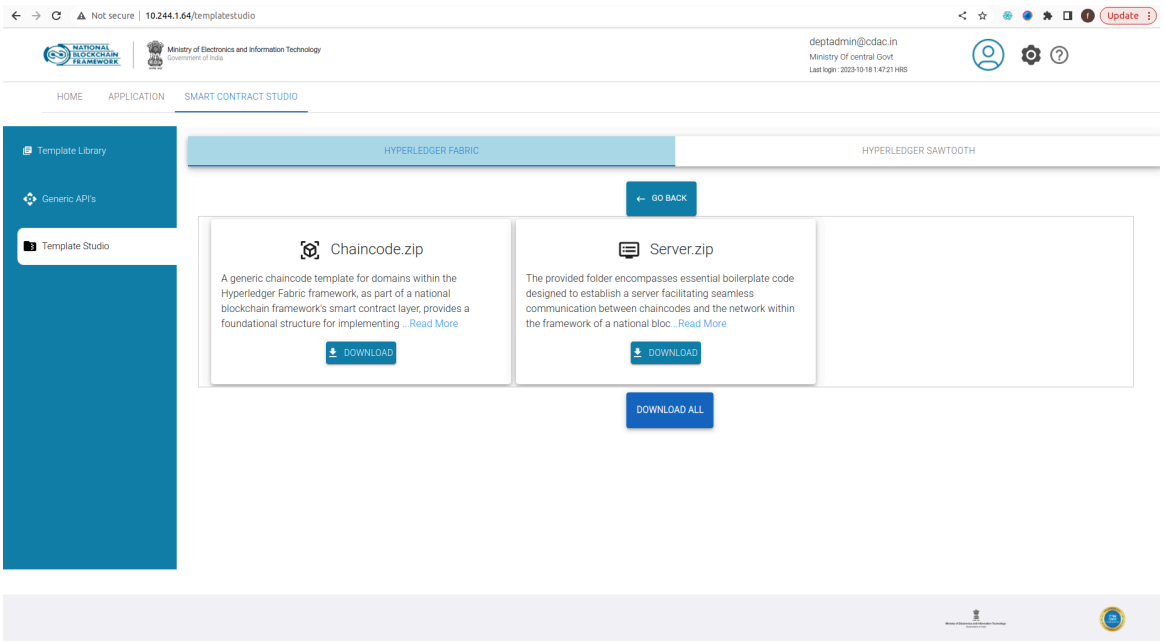


5. Template Studio

- Users can select language i.e Node or Go.User can select domain,based on the domain user can select stakeholders and functions and it also shows the number of functions selected for downloading and then user can download the chaincode bundled in a zip file. Here the chaincode downloaded is without asset management and design pattern.

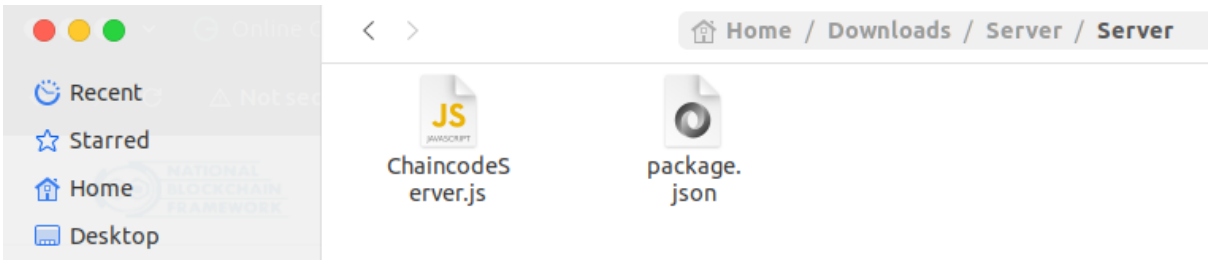


- The downloaded zipped file includes chaincode file and server file.



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- The Chaincode.zip will contain the chaincode in selected language whereas the server.zip when extracted will contains following files



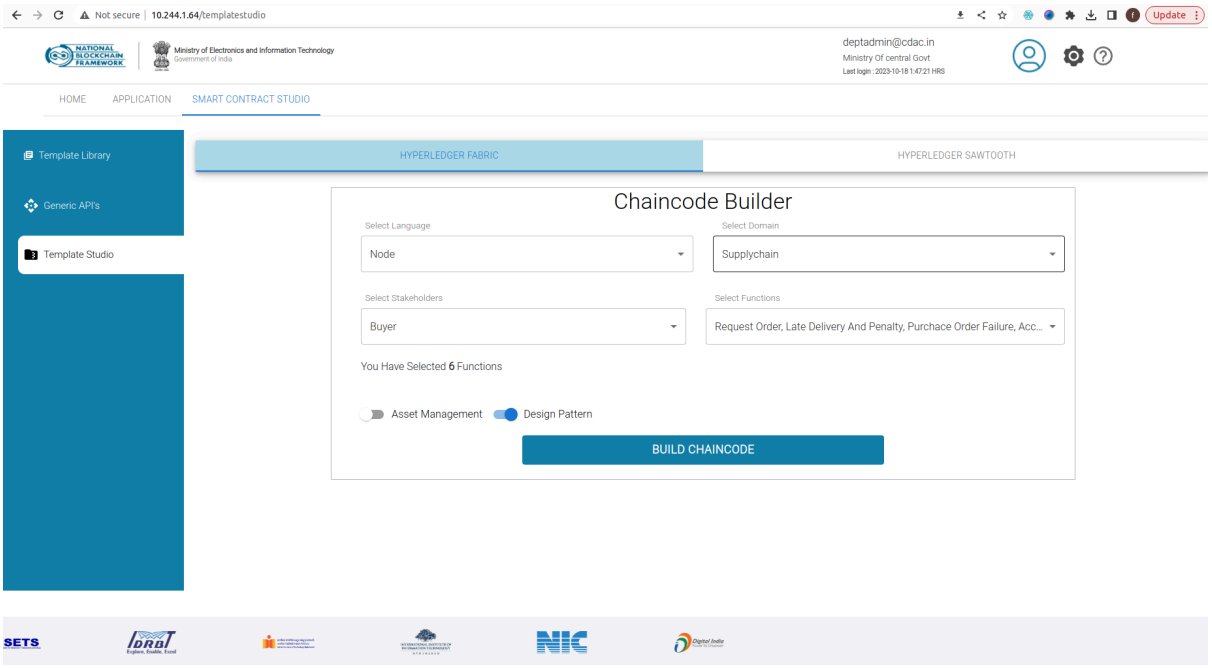
- **Before running the *ChaincodeServer.js* we need to modify the following configuration portion of the code in 2 Places according to *Sample Rest Server* IP and Port on which it's running.**
    - We need to change this configuration i.e IP and Port as per Deployed Sample Rest Server IP and Port.

```
13 app.use(bodyParser.urlencoded({ extended: false }));
14 app.use(cors());
15
16 //Server Configuration
17
18 // Note : Replace value for  apiUrl, token, cfgpath,  user, caname, msp, nodeType with sample rest server configurations
19 let apiUrl = "http://10.244.1.44:4000/fabric/v1/invokecc" //Configure Ip and Port
20 let token= "token"
21 let cfgpath = "/home/cdac/HLF-ubfagent/HLF-ubfagent/ubf/vars/profiles/sample_connection_for_nodesdk.json"
22 let user = "user1"
23 let msp = "cdac-com"
24 let nodeType = "true"
25 let args = []
26
27
28 app.post("/requestorder", (req, res) => {
29
30   let headers = {
31     'Content-Type': 'application/json',
32     'x-access-token': token,
33   }
34
35   if (Array.isArray(req.body)) {
36       // req.body is already an array, no conversion needed
37       args = req.body
38     } else if (typeof req.body === 'object' && req.body !== null) {
```

    - Also in the end of file we need to change this configuration i.e IP and Port as per Deployed Sample Rest Server IP and Port.

```
54
55
56   axios.post(apiUrl, data, { headers })
57     .then(response => {
58       // Handle success
59       console.log('Response:', response.data);
60     })
61     .catch(error => {
62       // Handle error
63       console.error('Error:', error);
64     });
65
66 });
67
68 app.listen(7000, "10.244.2.214"); // Configure Ip and Port for machine
69 console.log("Chaincode Server is  Running on http://10.244.2.214:7000");
70
71
```

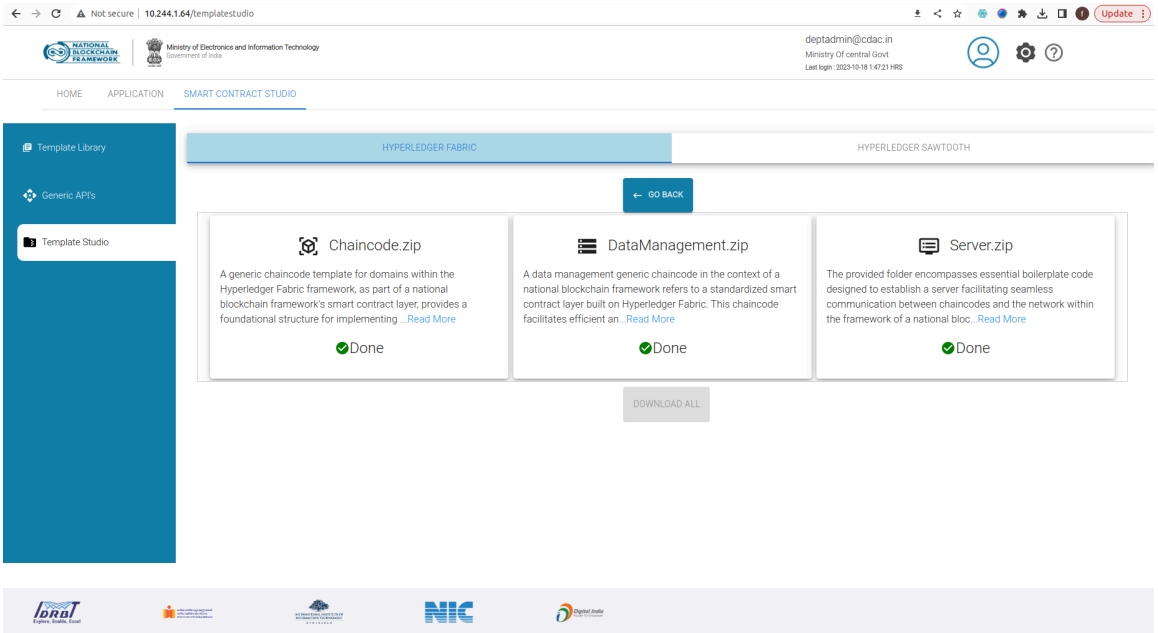
- If the user wants to include a design pattern along with the selected functions then the user can check the slider and can download the chaincode bundled in a zip file.



The downloaded file consists of **DataManagement.zip, Server.zip,** and **Chaincode.zip** files.

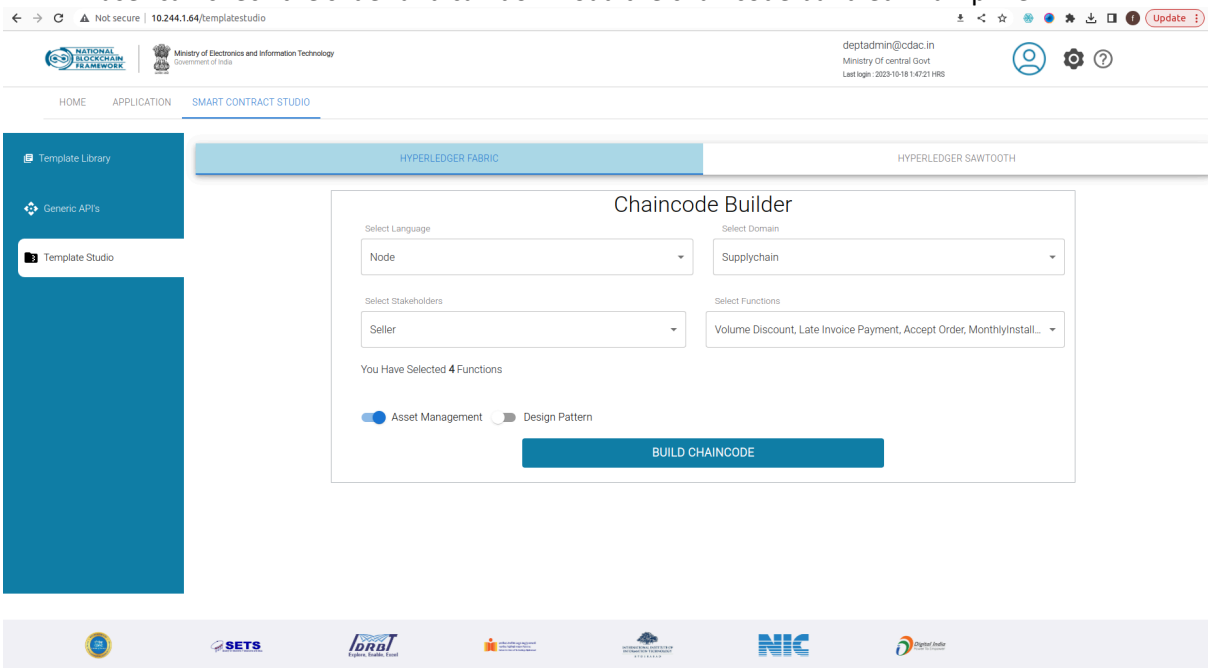
- **Chaincode.zip** contains the core logic of the chaincode.
- **DataManagement.zip** contains the generic data management part for the chaincode present in **Chaincode.zip**.
- **Server.zip** contains the boilerplate code for communication with Sample Rest Server

For smoother deployment of both chaincodes, it is recommended to **add "logic" and "data" as prefixes** while deploying them.

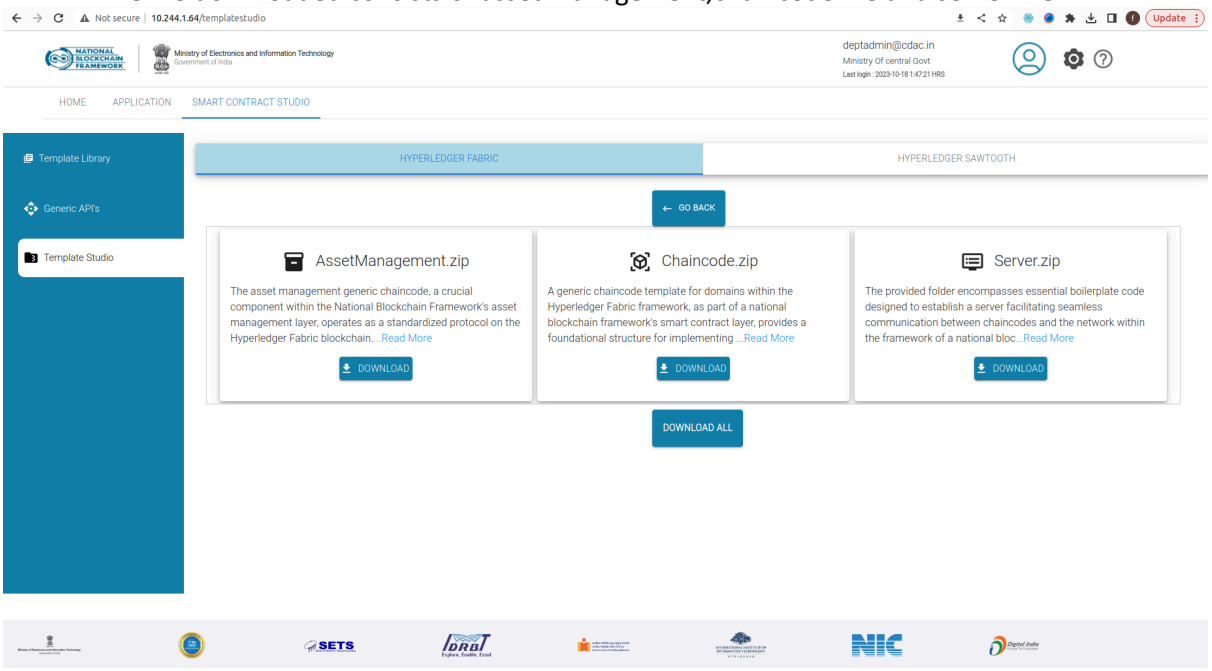


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- If the user wants to include asset management along with the selected functions then the user can check the slider and can download the chaincode bundled in a zip file.

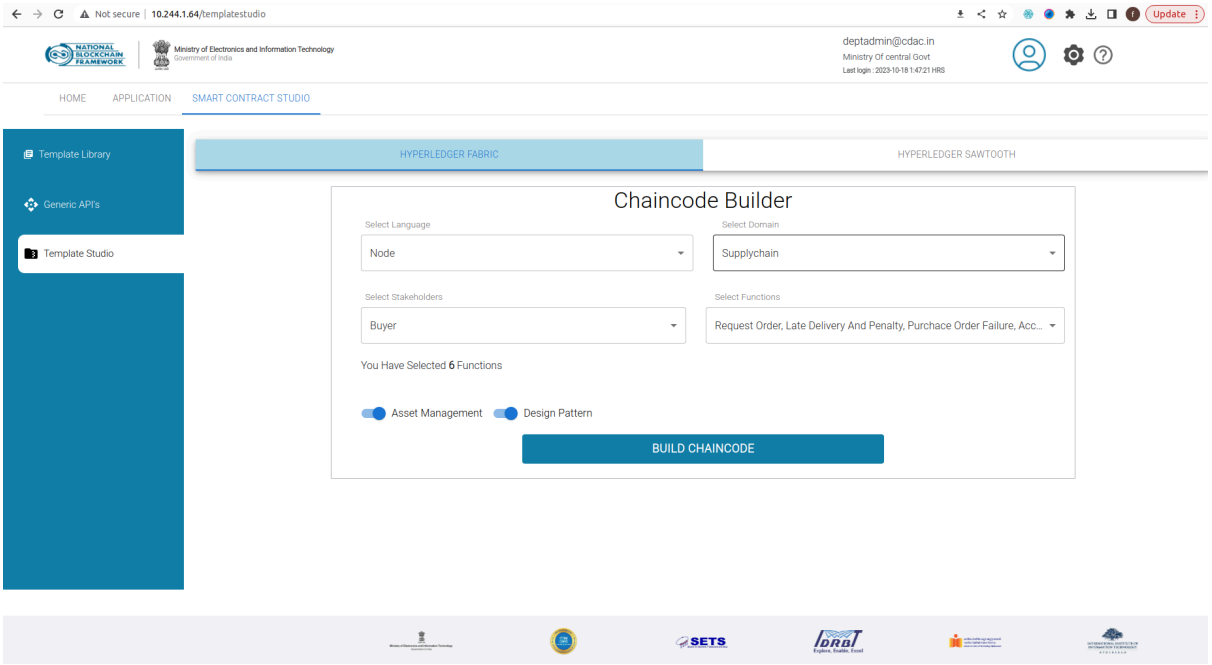


- The file downloaded consists of asset management,chaincode file and server file.

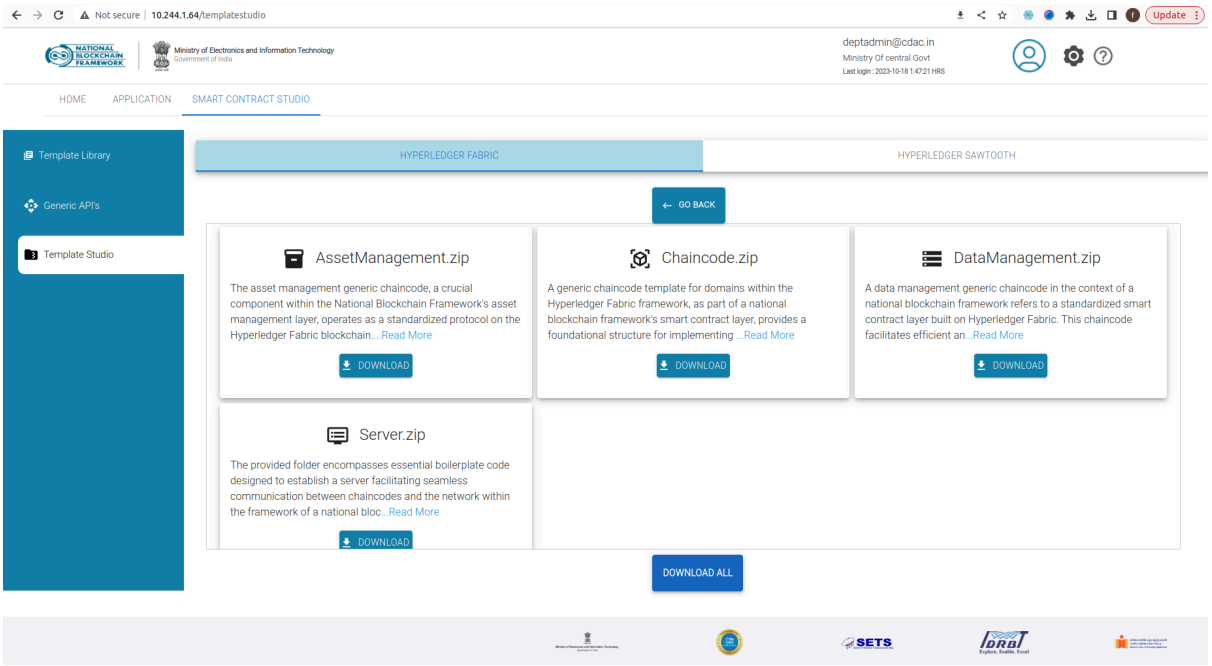


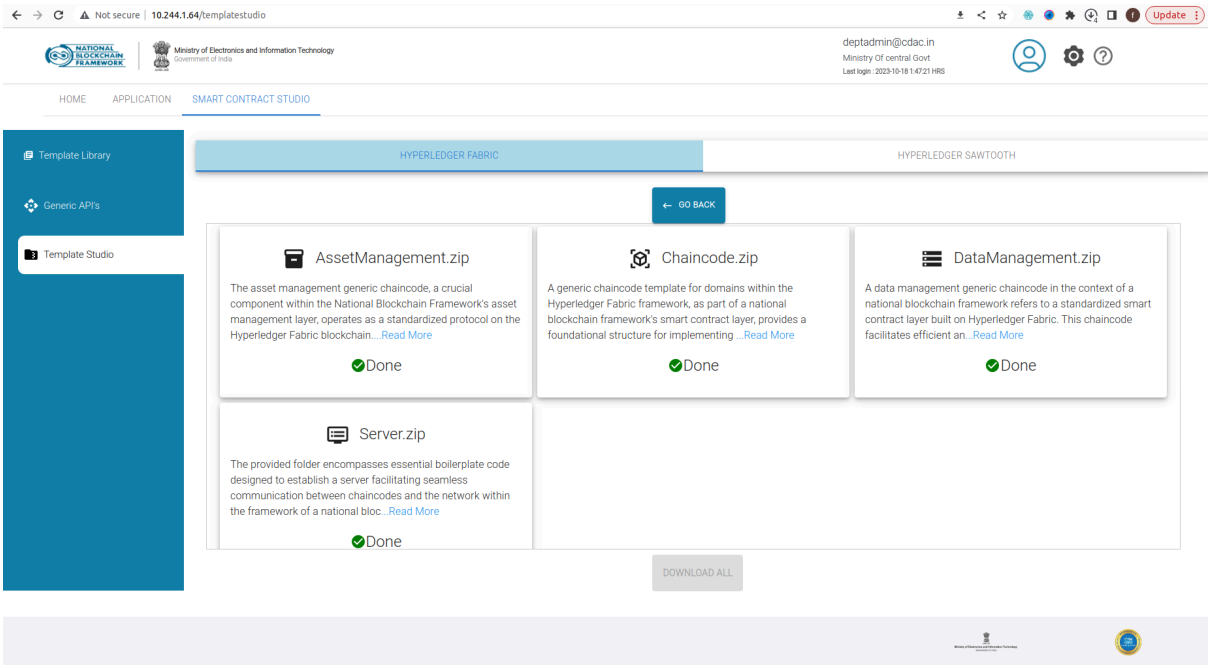
- **AssetManagement.zip** contains the core logic of the Asset Management Chaincode.

- If the user wants to include design pattern,asset management along with the selected functions then the user can check the slider and can download the chaincode bundled in a zip file.

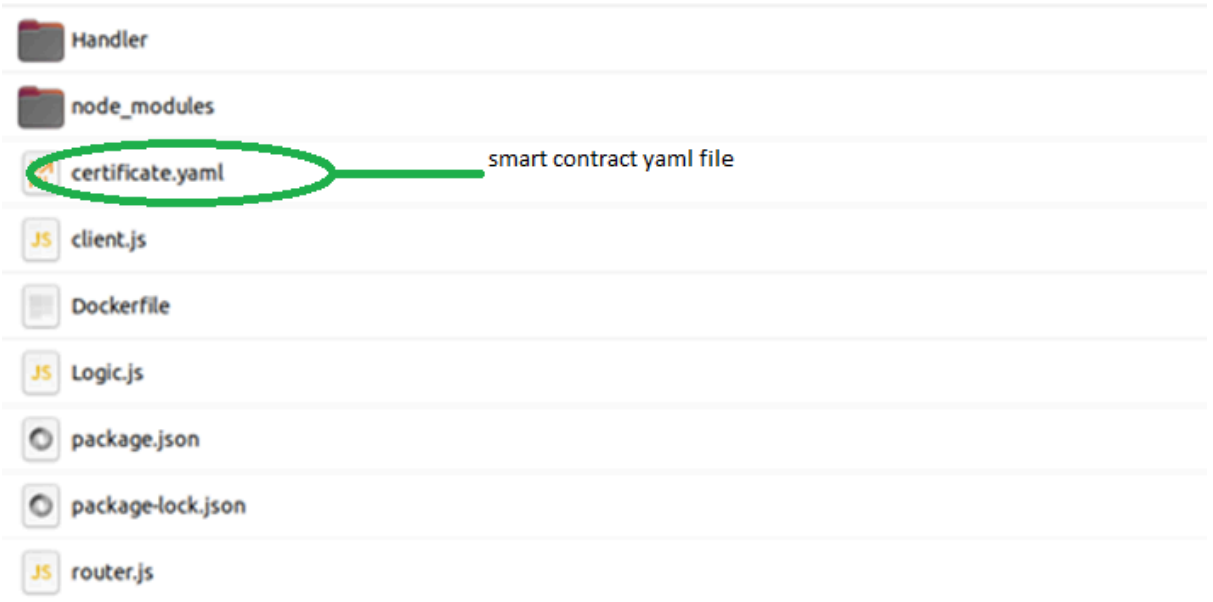


The downloaded zipped file consists of the following files.





- For Hyperledger Sawtooth platform downloaded zip file consist of following file:



If user wants to deploy downloaded chaincode in external Sawtooth network, then:

i) Edit client.js and Processor.js file(inside Handler folder) as per the directives mentioned in respective file's comment section.

ii) Keep network yaml file in the same path as that of smart contract yaml file.

6.Admin functionalities

Under the Admin page there are three functionalities provided.

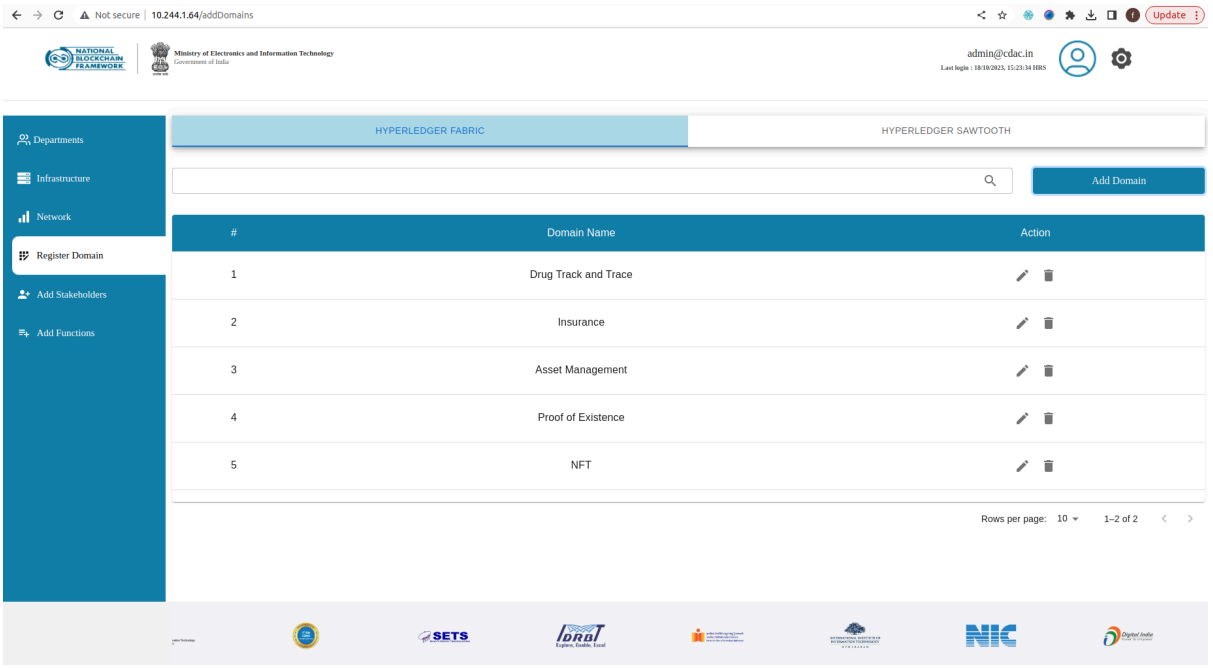
1.Register Domain

2.Add Stakeholders

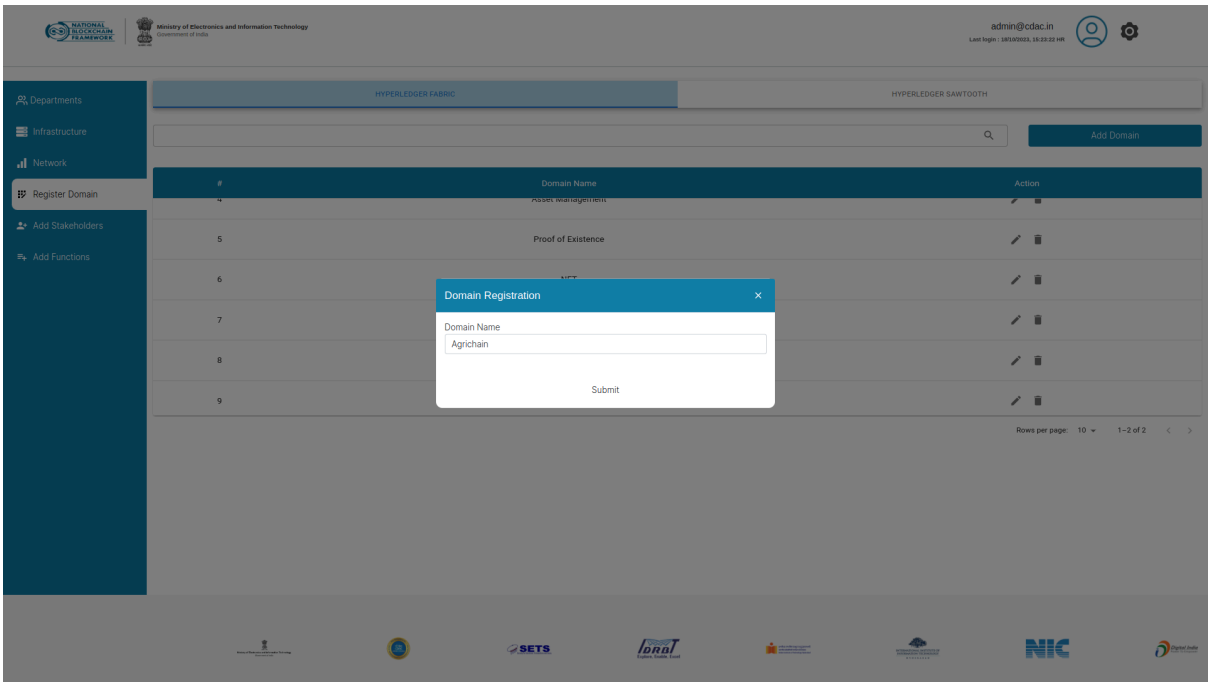
3.Add Functions

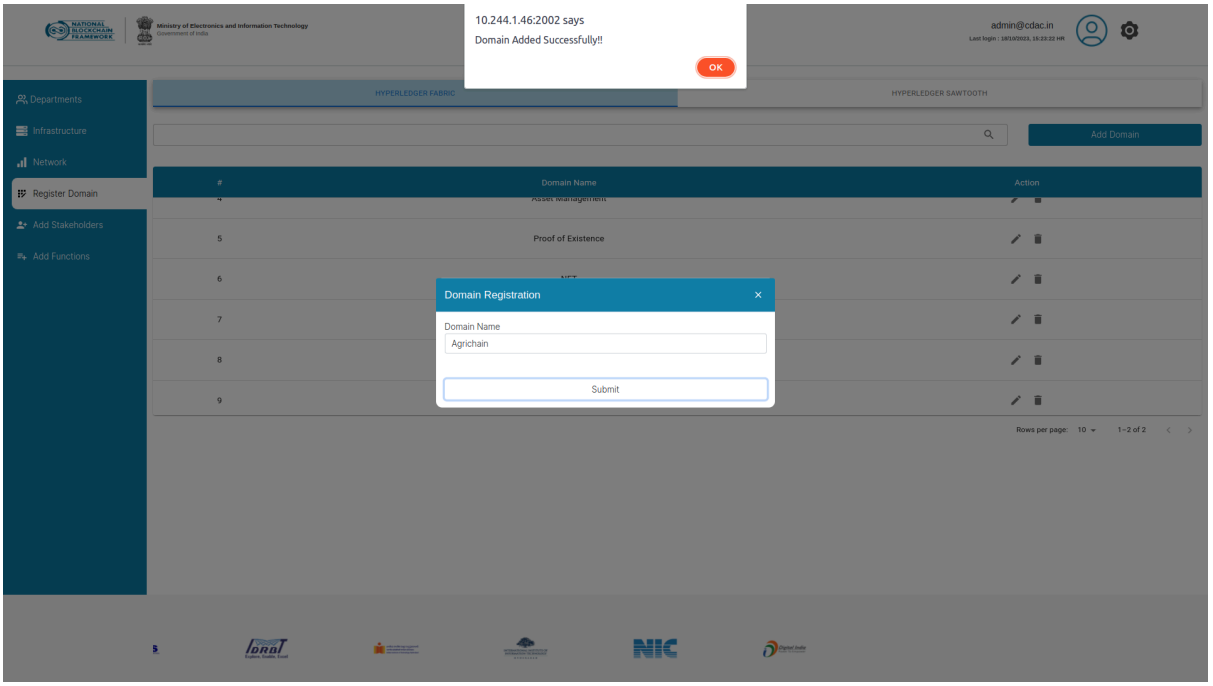
6.1 Register Domain

- Under the register domain tab,there is a provision of adding a new domain by clicking on “Add Domain” button.
- All the added domains will be displayed in a table.
- Admin can edit/delete the domain name.

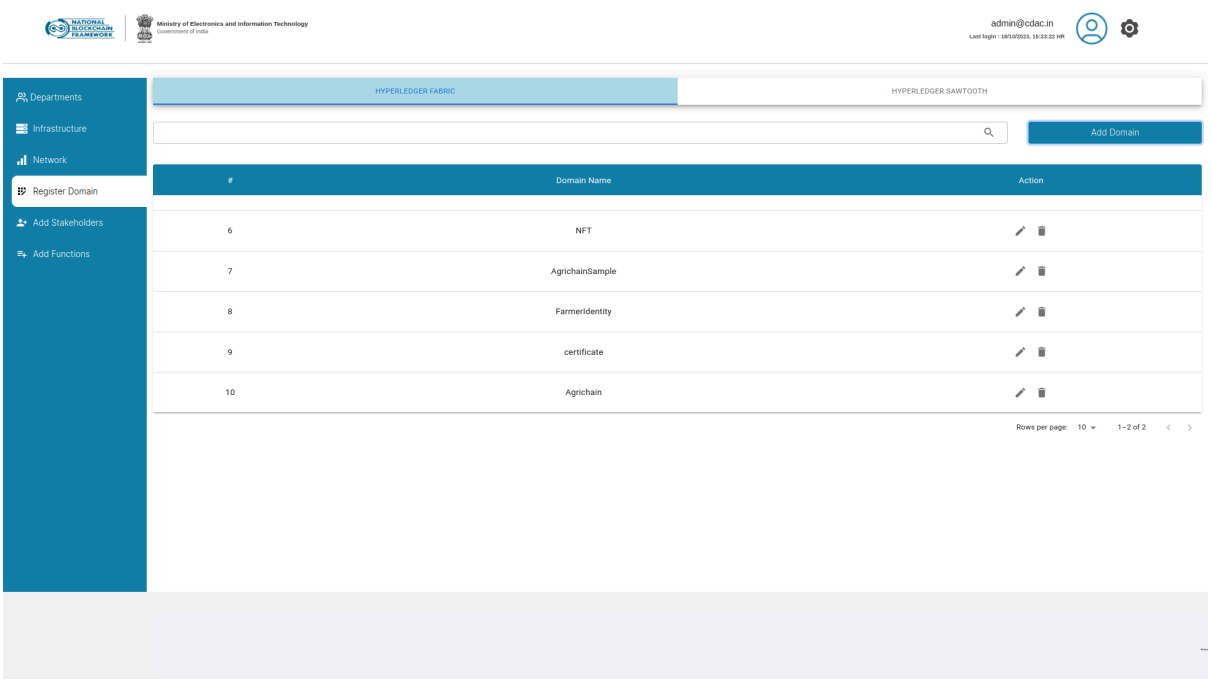


- Domain name Agrichain added by the admin.

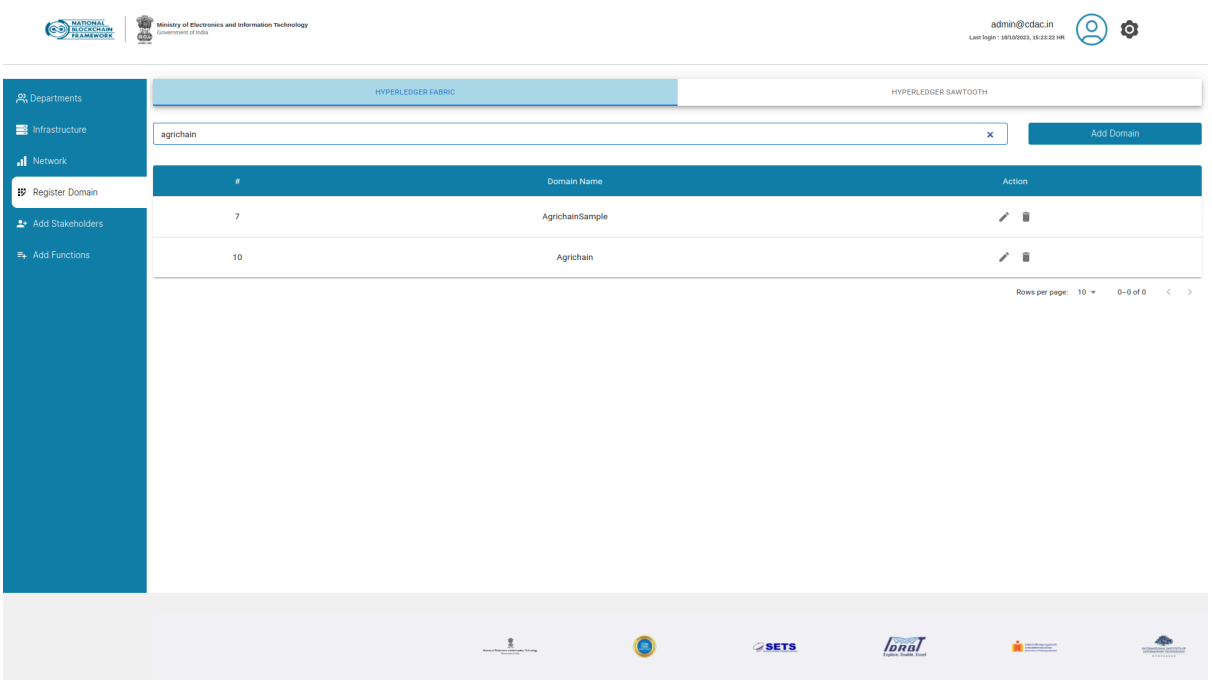




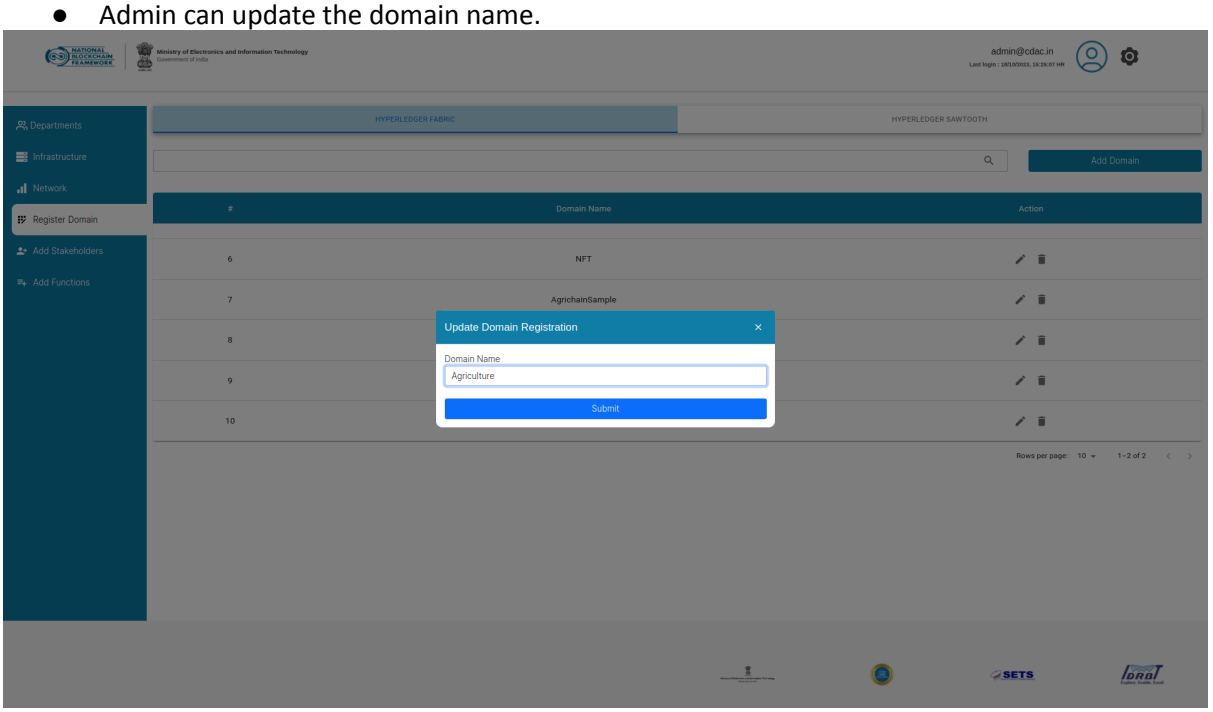
- Here,domain Agrichain was added succesfully.



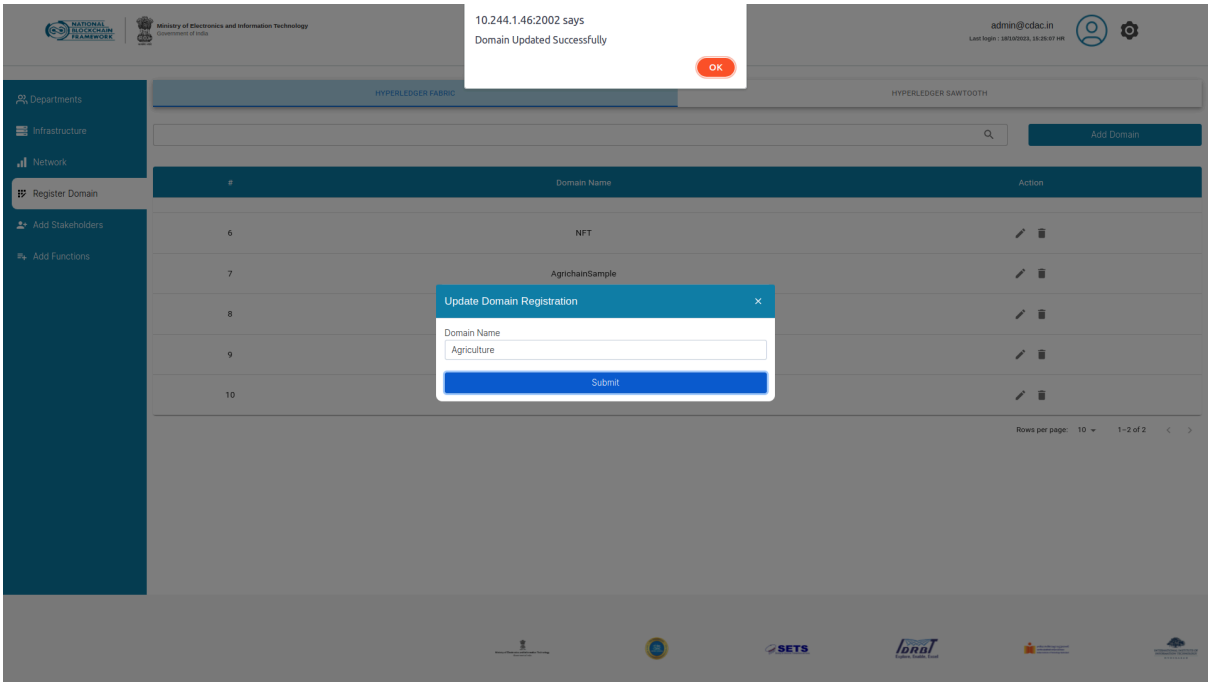
- Admin can search for the domain name also.



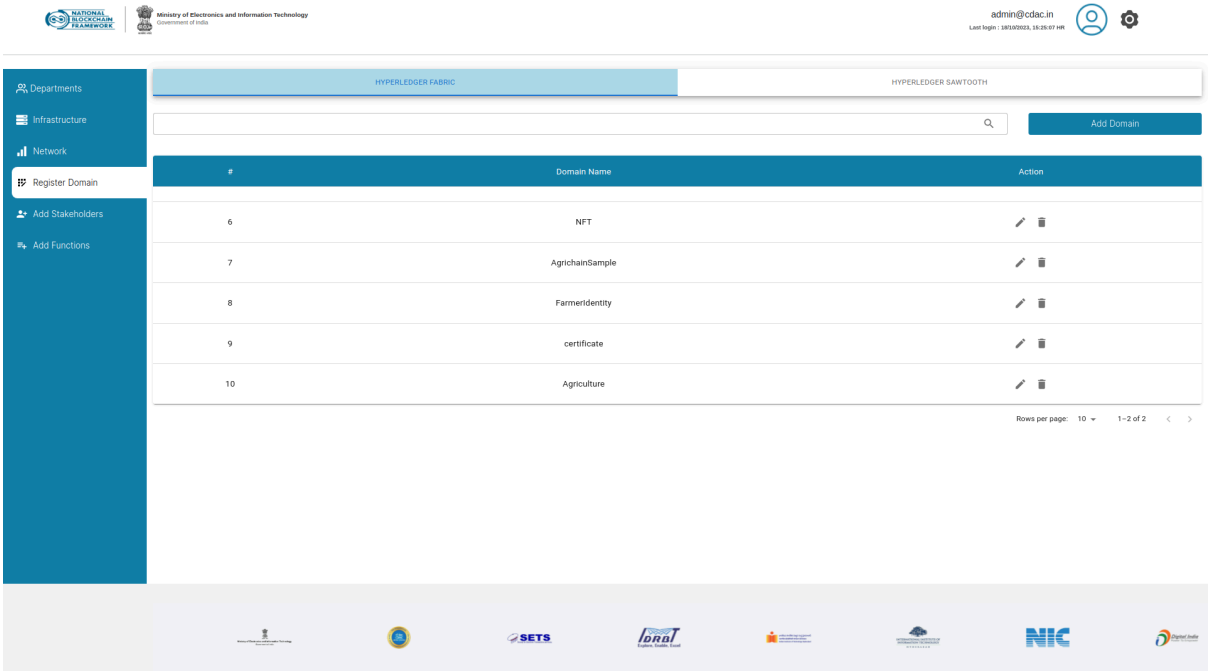
- Admin can update the domain name.



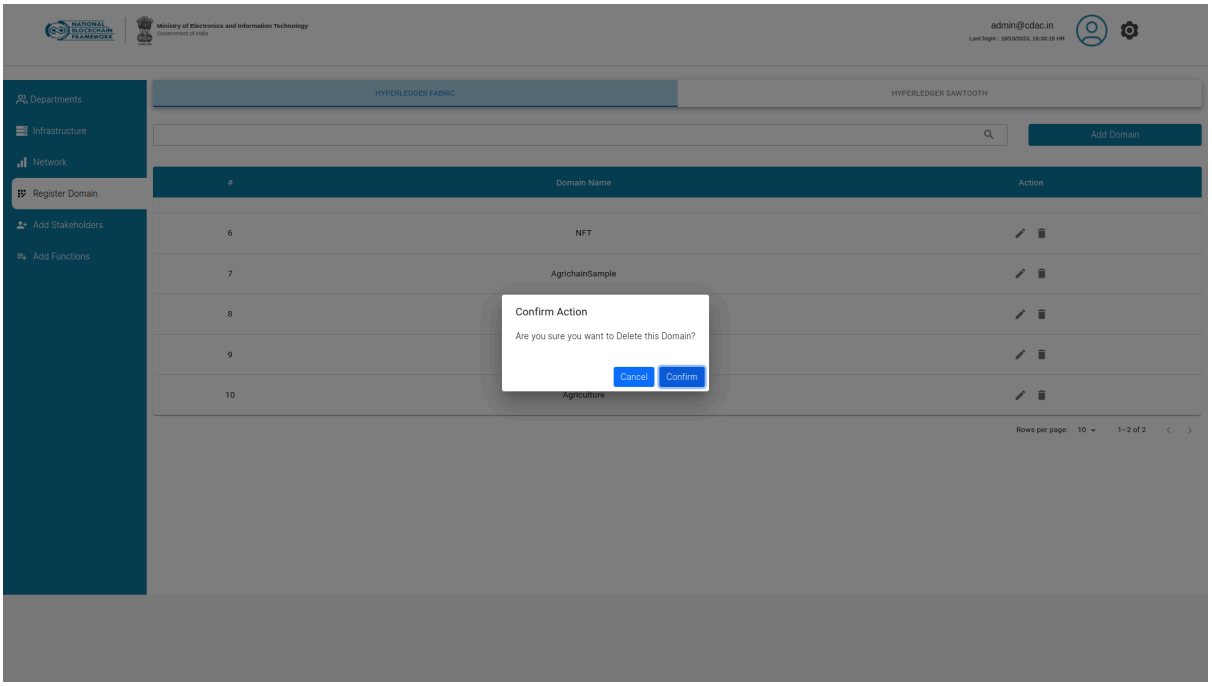
- Agrichain domain updated to agriculture domain successfully.



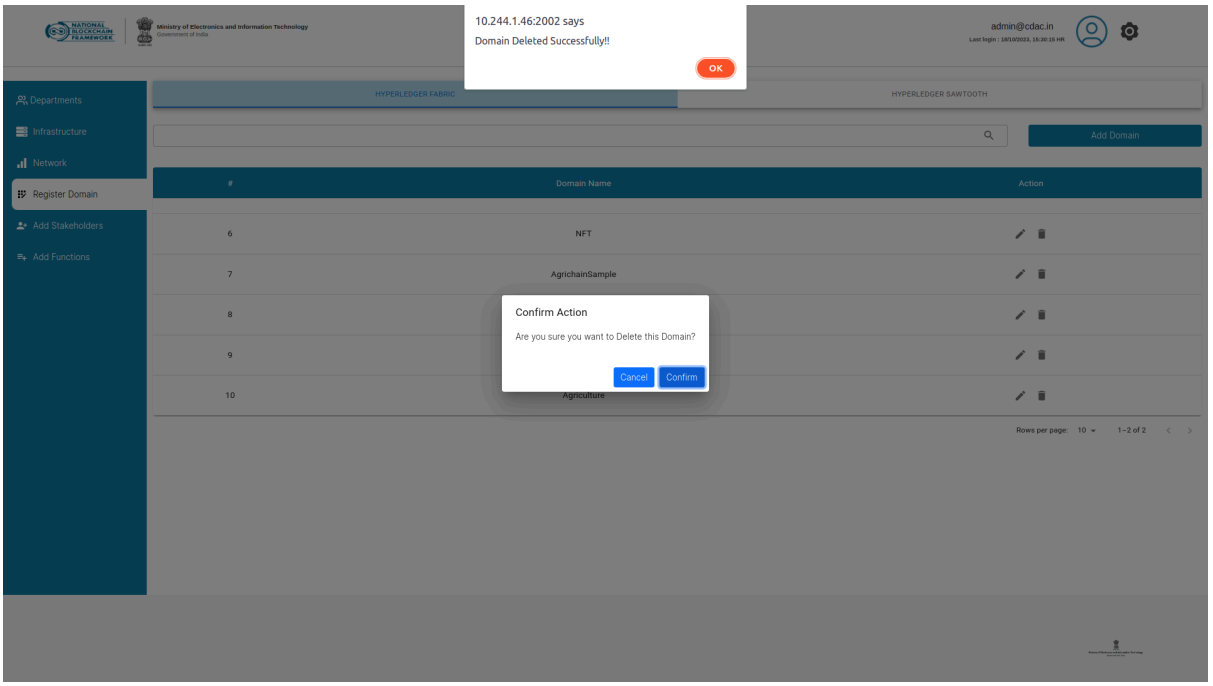
- Agrichain domain updated to agriculture displayed in the table.



- Admin can delete the domain.

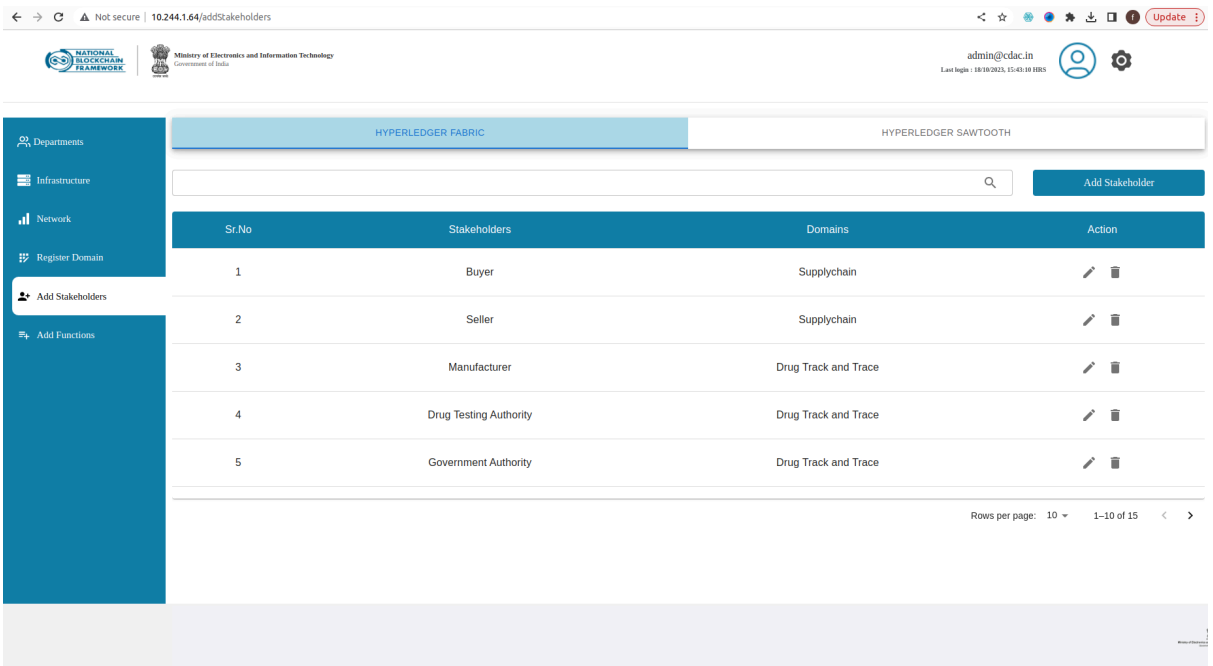


- Domain deleted successfully.

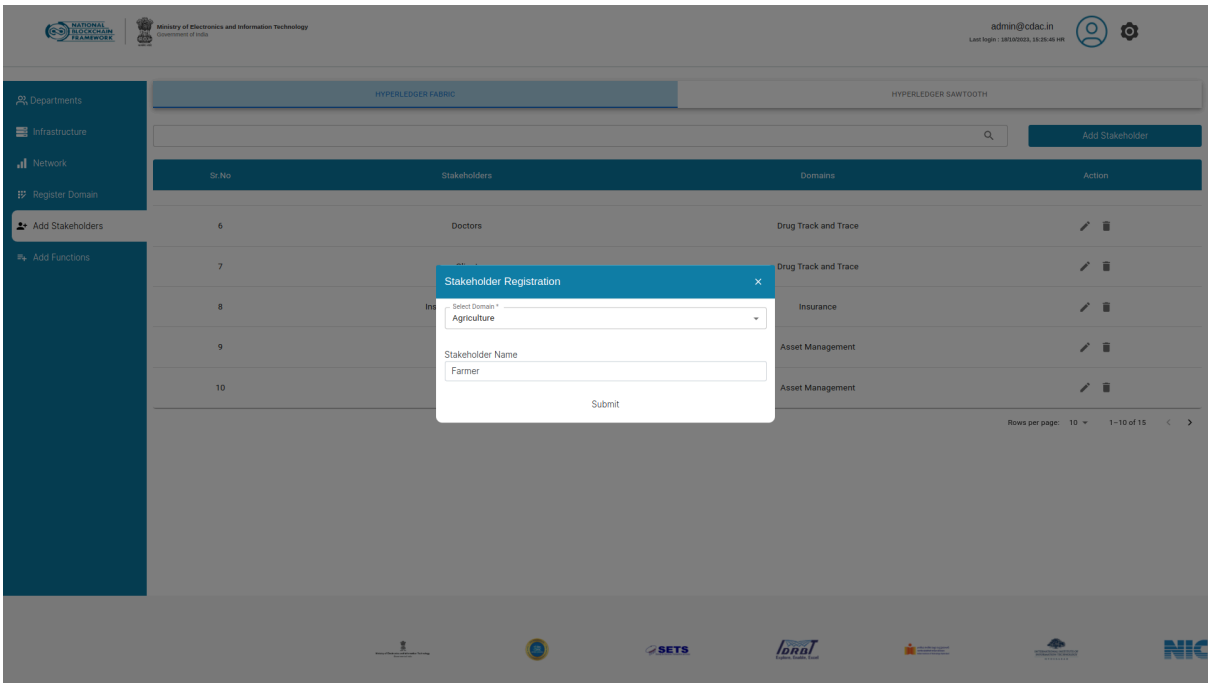


6.2 Add Stakeholders

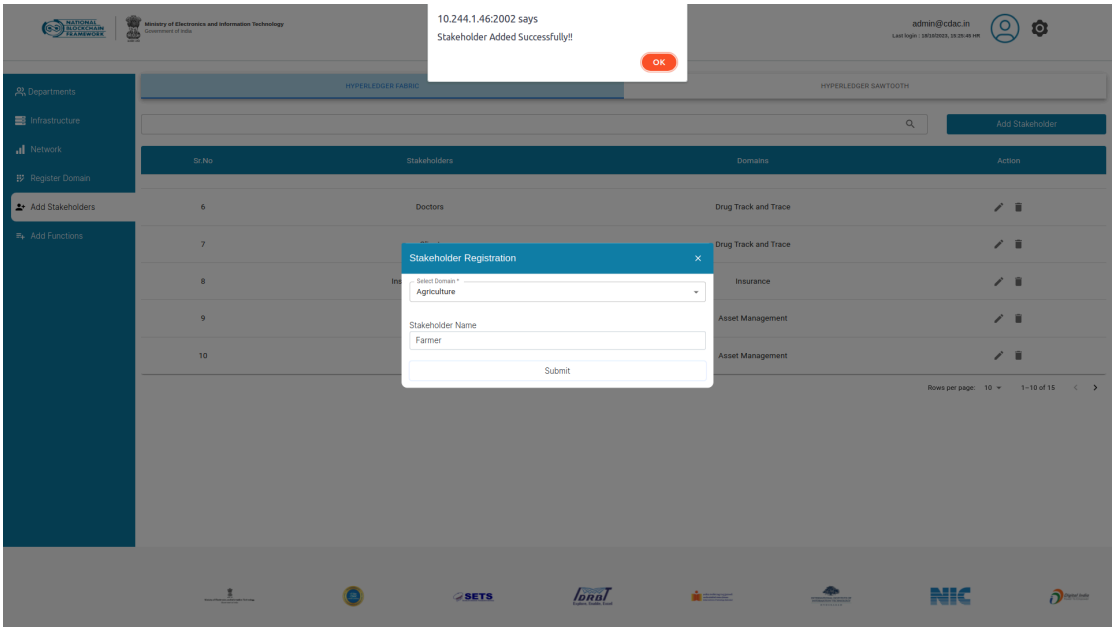
- Under the add stakeholder tab,there is a provision of adding a new stakeholder by clicking on “Add Stakeholder” button.
- All the added stakeholders will be displayed in a table.
- Admin can edit/delete the stakeholder name.
- Search facility is also provided to search for a particular stakeholder.

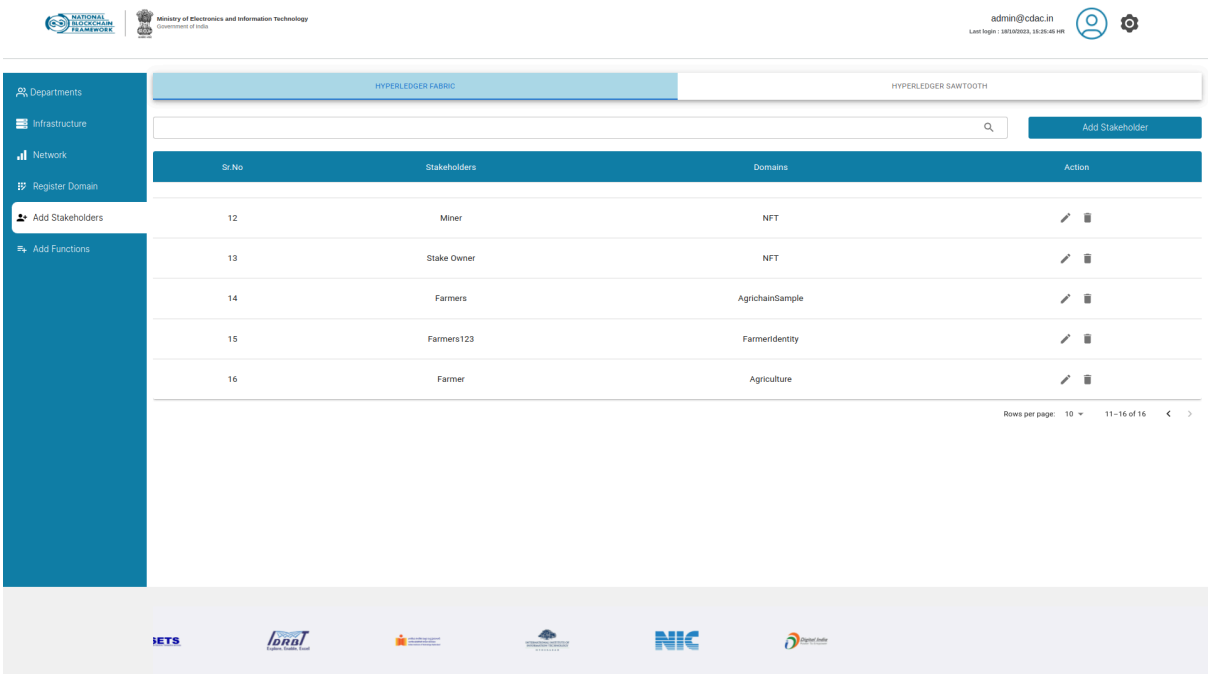


- Farmer stakeholder added under Agriculture domain

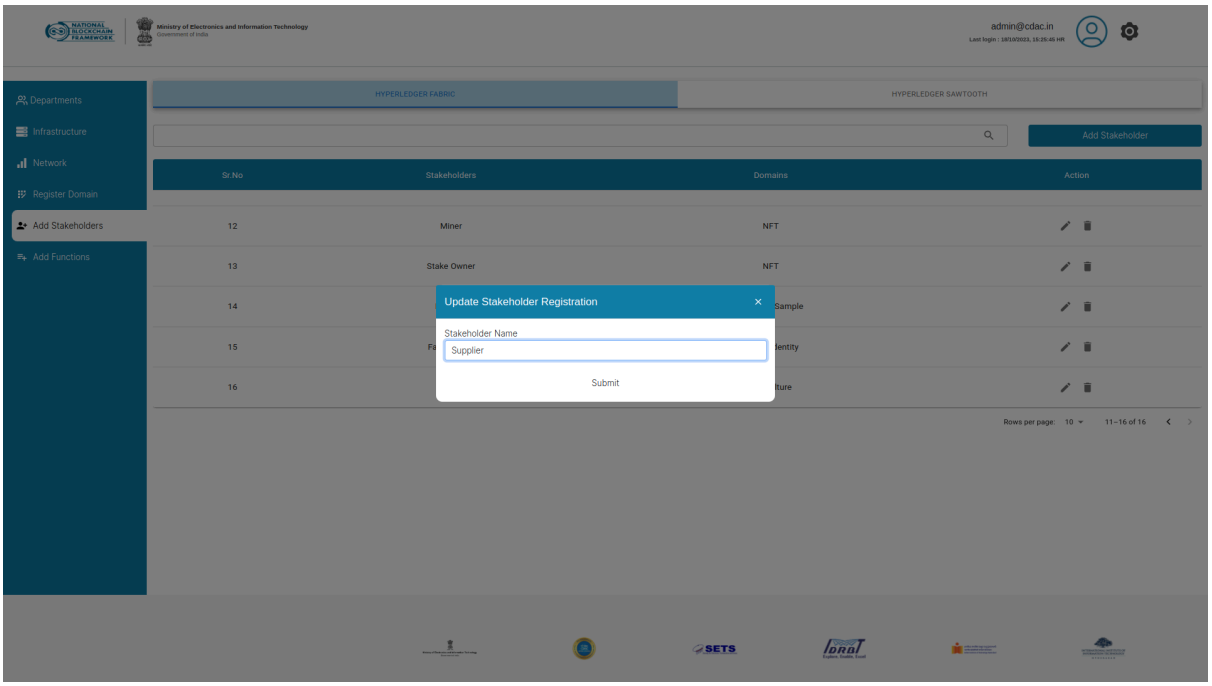


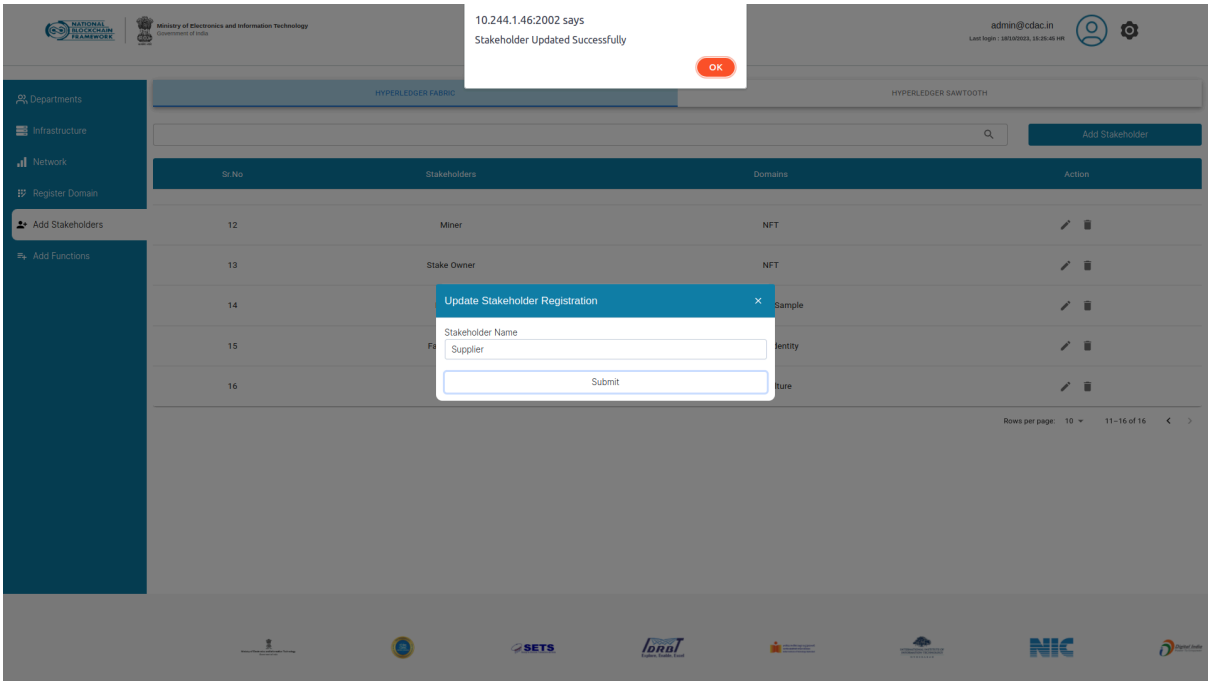
- Farmer stakeholder updated to supplier.

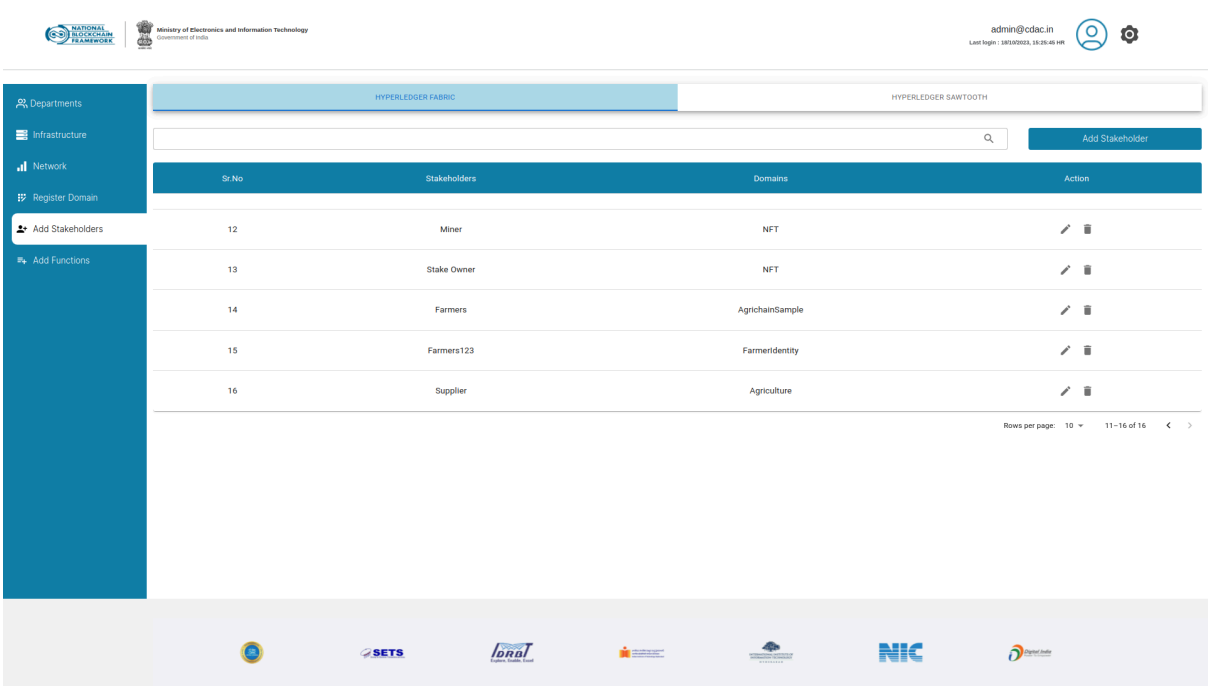




- Updated stakeholder shown in the table.



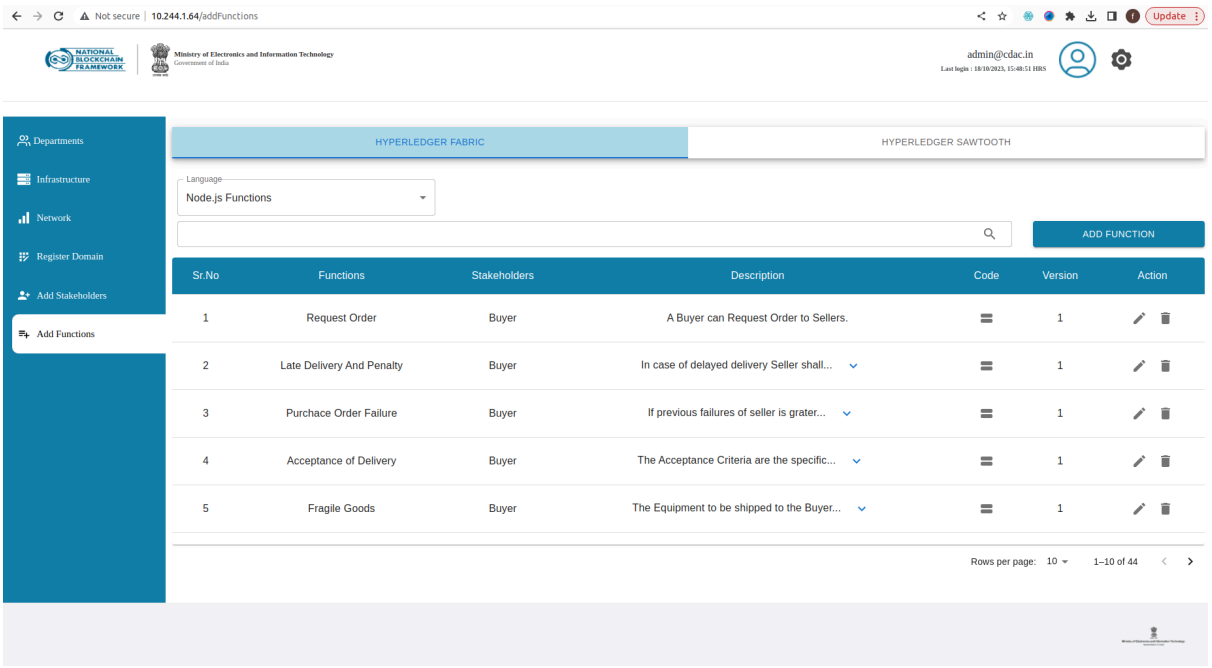




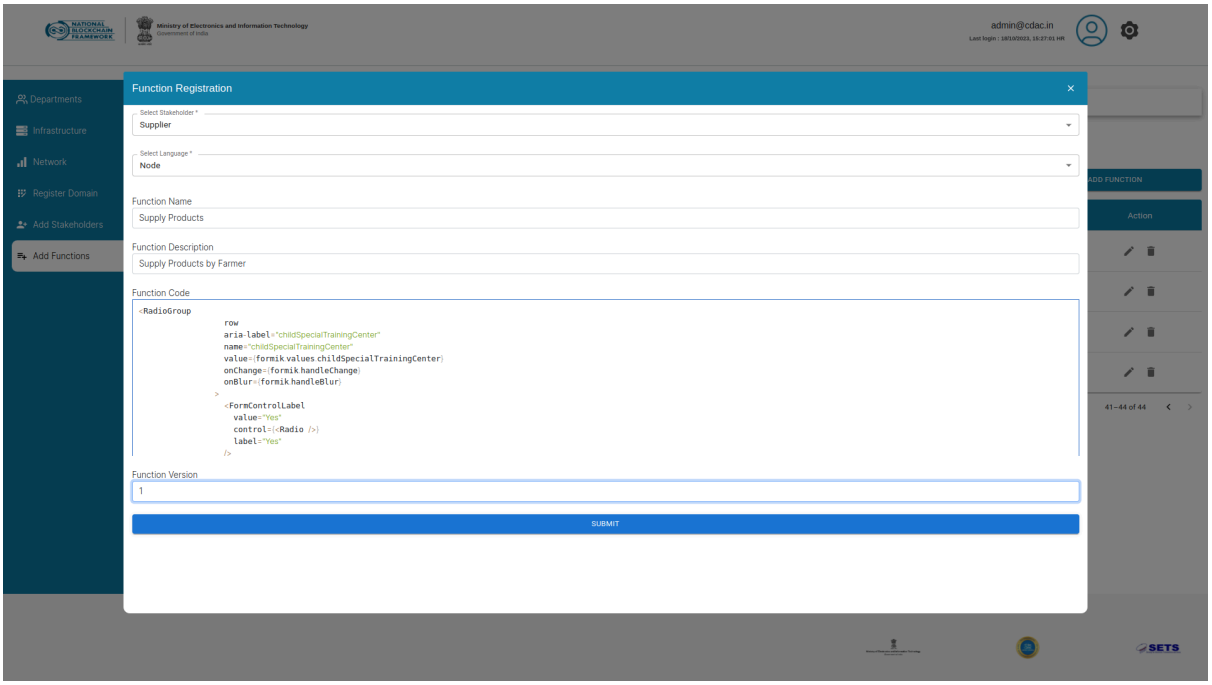
6.3 Add Functions-

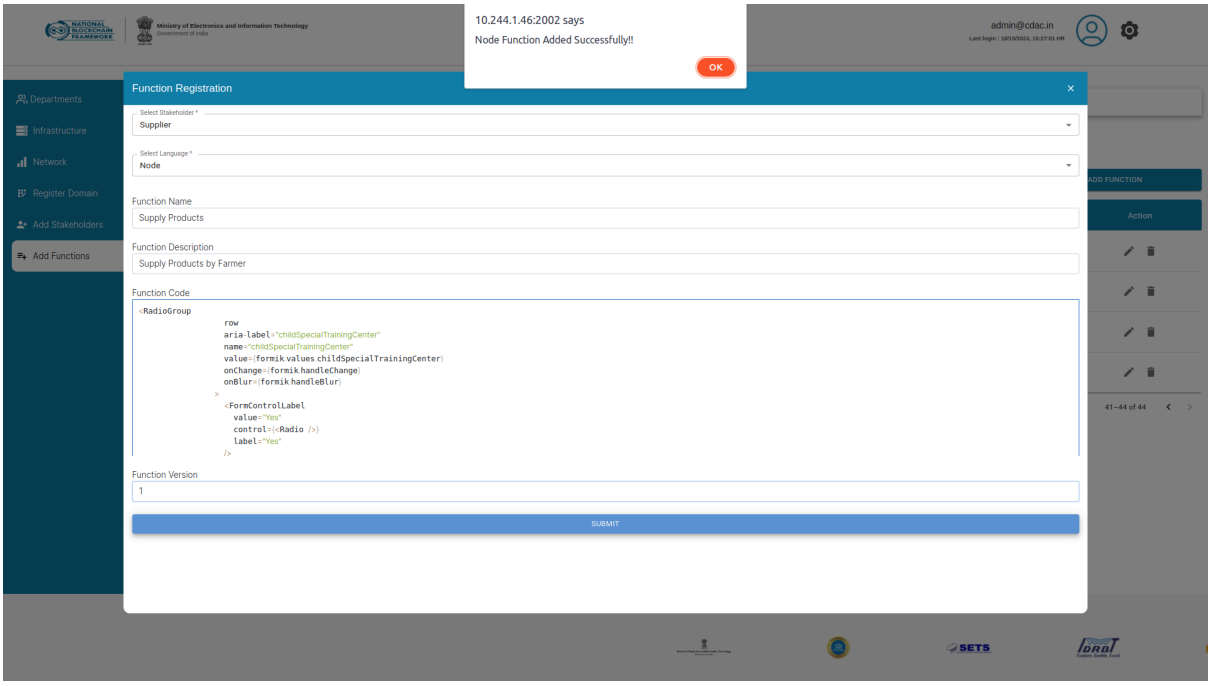
- Under the add Nodefunction tab and go function tab,there is a provision of adding a new function by clicking on the “Add Nodefunction or Add go function” button.
- All the added functions will be displayed in a table.
- Admin can edit/delete the function name.
- Search facility is also provided to search for a particular function.

NODE FUNCTIONS:

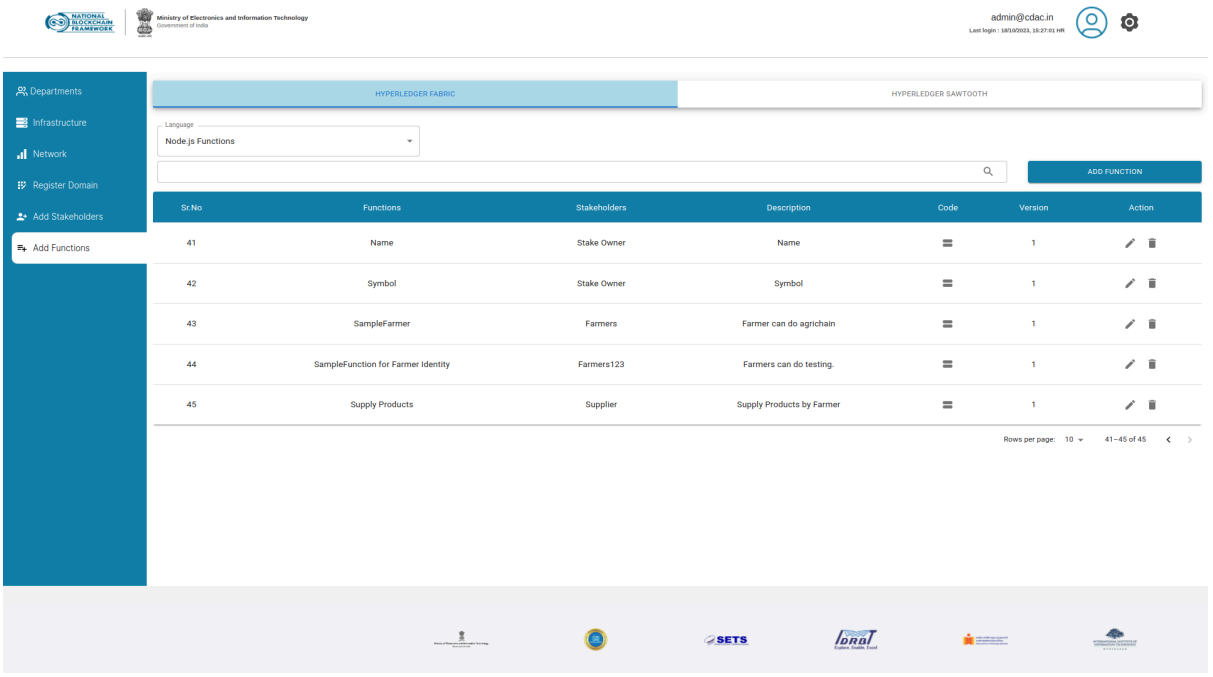


Function Name is added.



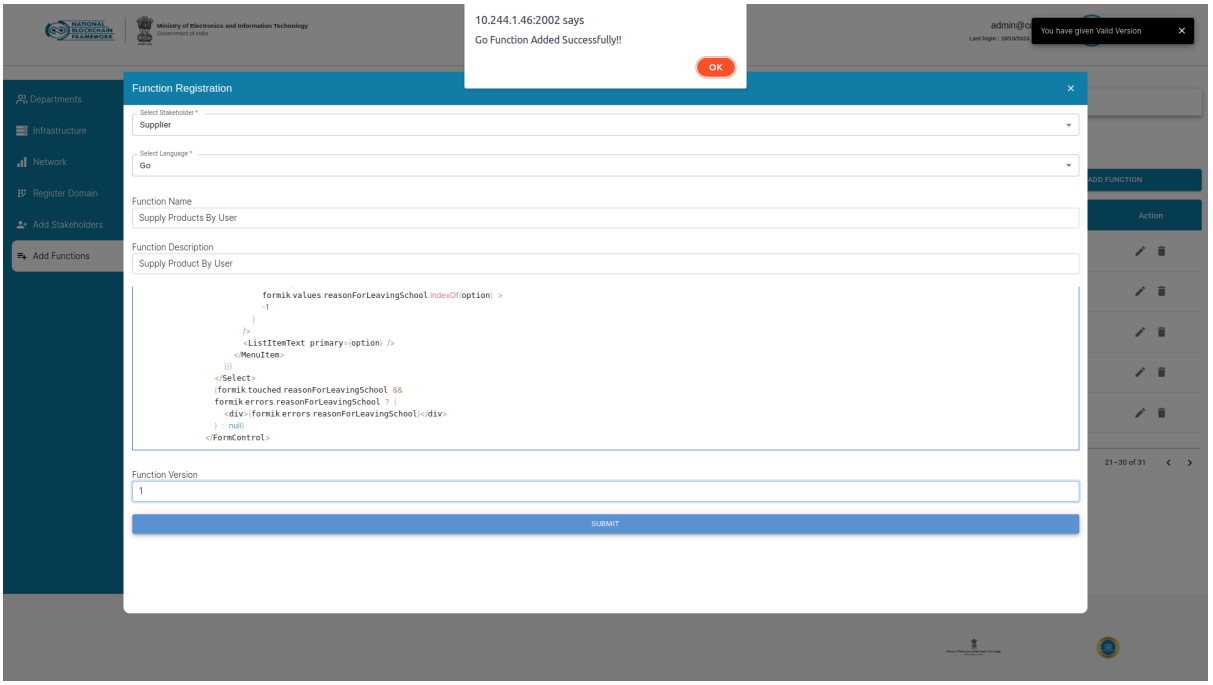


Function added is shown in the table.



GO FUNCTIONS:

Function is added successfully.



Added function is shown in the table.

